

Logistic regression: (supervised learning),

classification problem:

$$y \in \{0, 1\}$$

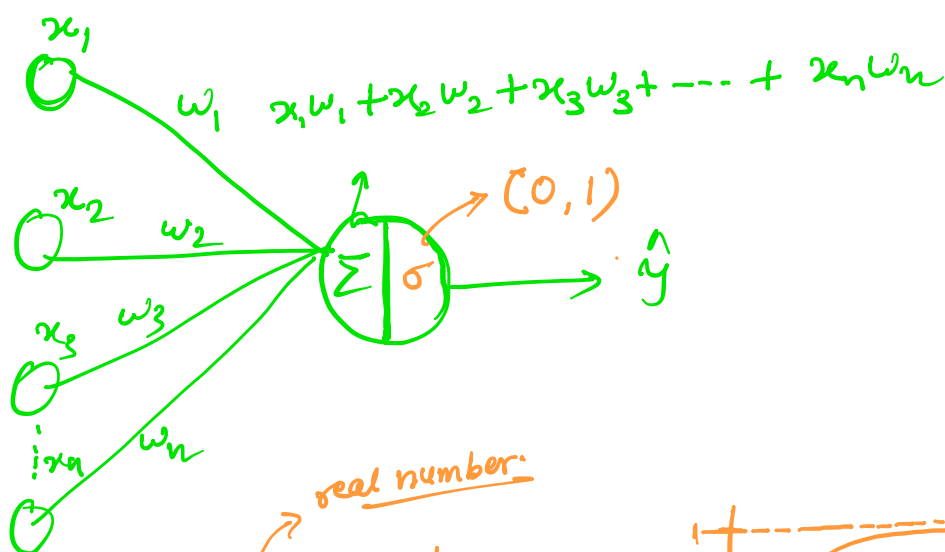
$$\hat{p} = P(y=1 | x)$$

conditional probability.

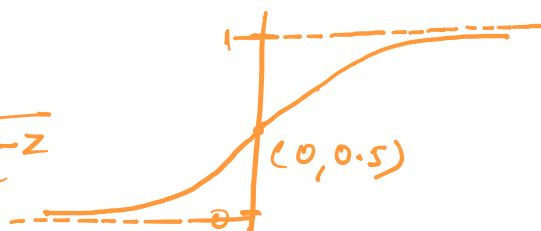
spam detection:  $y=1 \rightarrow$  spam  
 $y=0 \rightarrow$  not spam.

customer churn:  $y=1 \rightarrow$  customer leaves  
 $y=0 \rightarrow$  customer stays.

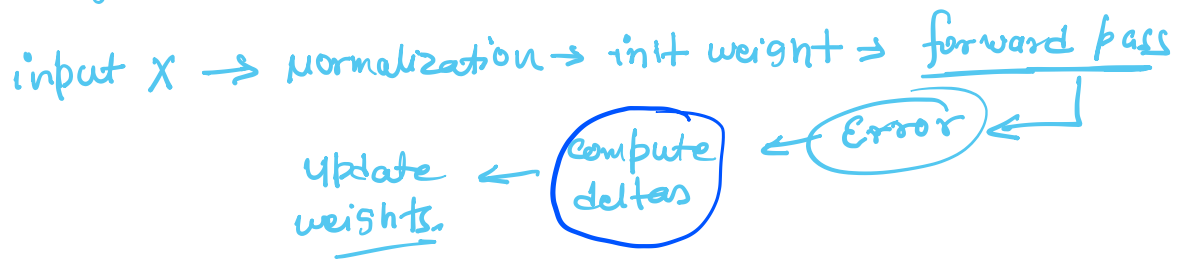
customer fraudulency:  $y=1 \rightarrow$  customer is fraud  
 $y=0 \rightarrow$  " " not fraud



real number:  
 $\sigma(z) = \frac{1}{1 + e^{-z}}$   
sigmoid function



Training pipeline:



$$\text{forward pass (estimation)} = \sigma(Xw)$$
$$(\hat{p})$$

threshold  $t$ .

$$\hat{y} = \begin{cases} 1 & \text{if } \hat{p} \geq t \\ 0 & \text{if } \hat{p} < t \end{cases}$$

$$x_0 = -3$$

$$w_0 = 1$$

$$x_1 = 8$$

$$w_1 = 0.04$$

$$x_2 = 1$$

$$w_2 = 1.2$$

$$\hat{p} = \sigma(x_0 w_0 + x_1 w_1 + x_2 w_2)$$

$$\hat{p} = \sigma(-3 \times 1 + 8 \times 0.04 + 1 \times 1.2)$$

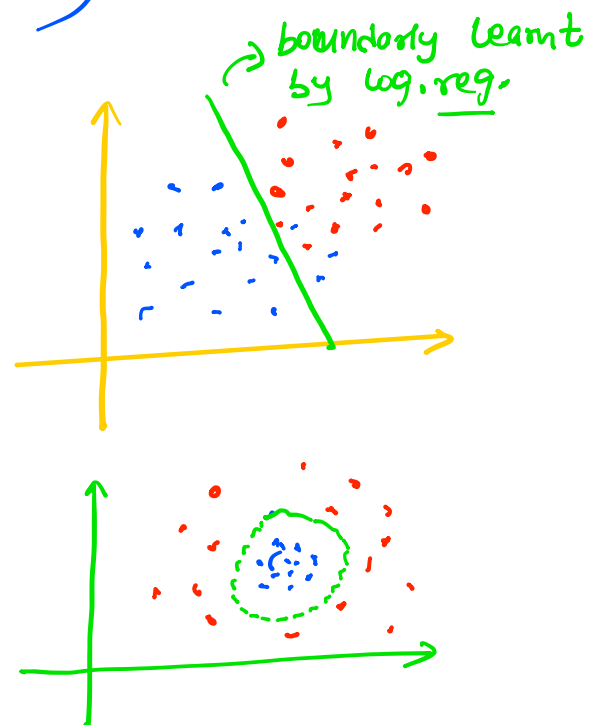
$$= \sigma(-3 + 0.32 + 1.2)$$

$$= \overset{\text{sigmoid}}{\sigma}(-1.48) \rightarrow z$$

$$\hat{p} = 0.185$$

$$t = 0.5$$

$$\hat{y} = 0$$



## Loss functions

when  $y = 1$

$$L = -\log(\hat{P})$$

when  $y = 0$

$$L = -\log(1 - \hat{P})$$

$$L(y, \hat{P}) = -[y \log(\hat{P}) + (1-y) \log(1-\hat{P})]$$

binary cross-entropy loss

$$J(w) = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{P}_i) + (1-y_i) \log(1-\hat{P}_i)]$$

negative log likelihood loss

$$P(y_i | x_i) = \hat{P}_i^{y_i} (1 - \hat{P}_i)^{1-y_i}$$

$$d(w) = \prod_{i=1}^n \hat{P}_i^{y_i} (1 - \hat{P}_i)^{1-y_i}$$

$$MLL(w) = \underbrace{-\log}_{\substack{\downarrow \\ \text{neg log}}} \underbrace{d(w)}_{\substack{\downarrow \\ \text{likelihood}}}$$

$$\frac{\partial J}{\partial w} = \{ ? \} \quad \frac{1}{n} x^T (\hat{P} - y)$$

$$w = w - \eta \cdot \frac{\partial J}{\partial w}$$

## metrics.

TN (true negative) :- pred = Neg. GT - Neg.

|    |   | prediction |    |
|----|---|------------|----|
|    |   | 0          | 1  |
| GT | 0 | TN         | FP |
|    | 1 | FN         | TP |

confusion matrix.

$y=0 \rightarrow$  no class

$y=1 \rightarrow$  class

15th  $\rightarrow$  class  $\rightarrow$   $\overset{y}{1}$   
 $\hat{y}=0$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

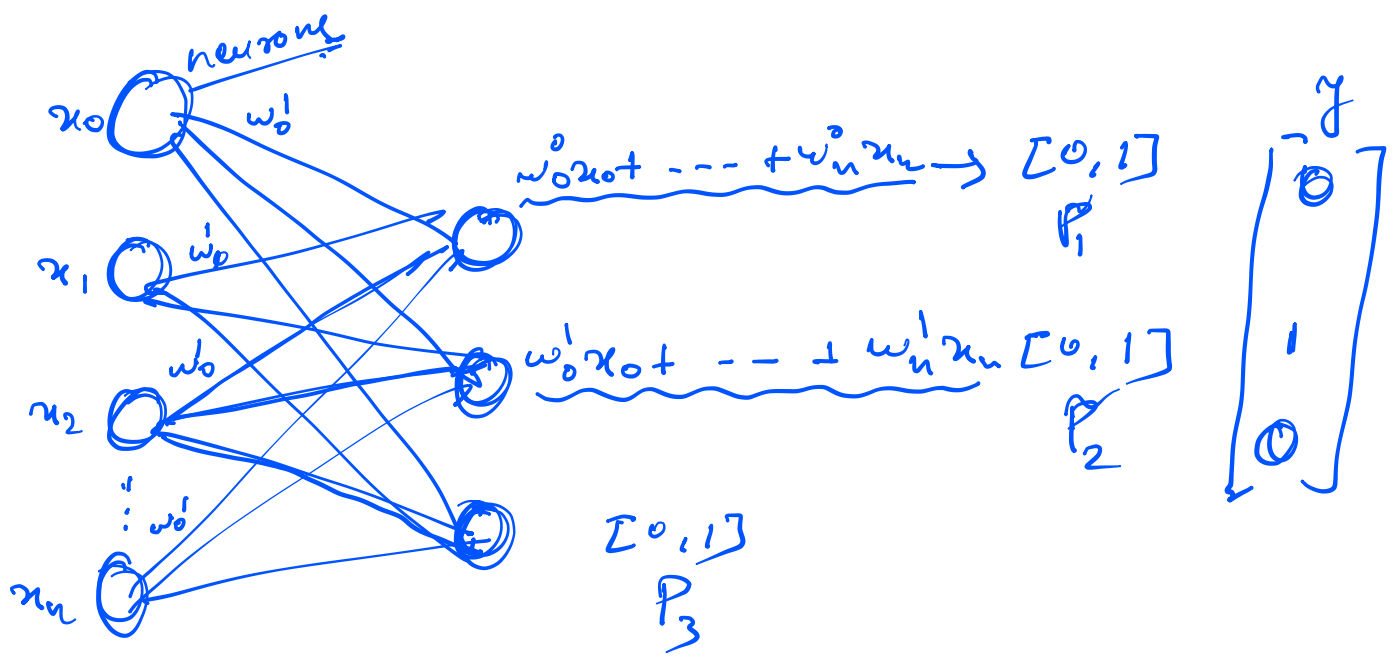
$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-score} = \frac{2 \cdot PR}{P + R}$$

$$t = 0.2$$

$\hat{p} = 0.3$

|           |                |
|-----------|----------------|
| 0         | 1              |
| no cancer | <u>cancer.</u> |



$$p_1 + p_2 + p_3 = 1$$